

FAHIM SHAHRYER SIZAN

Undergraduate Software Engineering Student

01828624005 | bsse1531@iit.du.ac.bd | Fazlul Huq Muslim Hall, University of Dhaka

github.com/Sizan008 | linkedin.com/in/fahim-shahryer-sizan

Profile

Software Engineering undergraduate at the Institute of Information Technology, University of Dhaka, with hands-on experience in full-stack web and mobile application development. Experienced with Angular, Spring Boot, Flutter, FastAPI, REST API integration, relational databases, Firebase, and basic machine learning workflows. Interested in building secure, maintainable, and user-focused software systems.

Education

Bachelor of Science in Software Engineering (Undergoing)	2023–2027 (Expected)
Institute of Information Technology, University of Dhaka	CGPA: 3.14/4.00 (up to 5th semester)
Higher Secondary Certificate (HSC)	2022
Bogura Government College	GPA: 5.00/5.00
Secondary School Certificate (SSC)	2020
Cantonment Public School and College, Saidpur	GPA: 5.00/5.00

Technical Skills

Programming Languages	Python, Dart, Java, C++, C
Frontend	Angular, HTML, Tailwind CSS, SCSS
Backend	Spring Boot, FastAPI, REST API Integration
Mobile / Realtime	Flutter, Firebase Realtime Database
Databases	PostgreSQL, MySQL
Machine Learning	Keras, Basic Model Training and Inference
Computer Vision	OCR, face-api.js, Face Detection, Face Verification, Liveness Detection
Other	Git/GitHub, Software Requirements Documentation, Team Collaboration, Communication

Selected Projects

Management Console – Angular & Spring Boot 2026

- Worked on an enterprise management console with Angular frontend and Spring Boot based backend REST APIs.
- Developed and integrated modules for user management, authorization, workflow, configuration, reporting, and other administrative functionalities.
- Implemented search, filtering, pagination, responsive forms, validation, reusable UI components, and API-driven data loading.
- Worked with frontend-backend integration, authentication-related workflows, database-driven APIs, and maintainable modular architecture.

eKYC Digital Account Opening System 2026

- Developed an Angular-based digital account-opening workflow covering product selection, branch selection, OTP verification, customer information, nominee details, signature, and account opening.
- Implemented front and back NID image upload with OCR-based extraction and verification of customer identity information.
- Used `face-api.js` for browser-based face detection and facial verification during the customer onboarding process.
- Worked with liveness detection, step-based validation, backend API integration, and controlled progression through the complete eKYC onboarding flow.

IAmFertilizer – Smart Fertilizer Recommendation for Paddy 2025–2026

- Developed a mobile application that analyzes rice-leaf images to classify plants as Healthy or detect Nitrogen (N), Phosphorus (P), and Potassium (K) deficiencies.
- Integrated the trained image classification model with a FastAPI backend to generate targeted fertilizer recommendations based on the detected nutrient condition.
- Added mobile image capture and history features for field use. Tech Stack: Python, Dart, Flutter, FastAPI, Keras, PostgreSQL.
- [GitHub Repository](#)

Connect-4 Android Game

2025

- Built an interactive Connect-4 mobile application using Flutter and Dart with Player vs Player and Player vs AI modes.
- Implemented multiple AI difficulty levels using the Minimax algorithm with Alpha-Beta pruning for efficient decision making.
- Added animations, responsive game interactions, and a clean mobile UI for a polished gameplay experience.
- [GitHub Repository](#) | [APK](#)

BokBok – Group Chat Application

2026

- Developed a Flutter-based realtime group chat application using Firebase Realtime Database.
- Implemented authentication, group creation and joining, member management, group search, and realtime messaging.
- Used live Firebase updates to synchronize messages and group-related information between users instantly.
- [GitHub Repository](#) | [APK](#)

Automatic Car Parking Using Arduino

2024

- Designed a prototype for automatically monitoring parking-lot availability using Arduino and ultrasonic sensors.
- Integrated LEDs and an LCD screen to detect vehicle presence and display the realtime number of available parking spaces.
- [Project Demo](#)

Chor-Dakat-Police-Babu Game

2024–2025

- Developed a digital version of the traditional Chor-Dakat-Police-Babu game using C++ and SFML.
- Implemented single-player gameplay with computer-controlled roles as well as multiplayer support for four human players.
- Added random role assignment and round-based gameplay focused on deduction, strategy, and luck.
- [GitHub Repository](#)

DU Food Point Management System – SRS

2025

- Co-authored a detailed Software Requirements Specification for a food point management system.
- Worked on requirement analysis, functional requirements, system behavior, and overall software specification documentation.
- [SRS Document](#)

Tutoring & Volunteering

- Worked as a Physics and Mathematics tutor at Optimum Teaching Center, Farmgate.
- Privately tutored more than 30 students.
- Volunteered in school cultural and sports events.

Co-Curricular Activities

- Represented IIT Cricket Team in the Inter-Department University Tournament.
- Represented IIT Football Team in the Inter-Department University Tournament.
- Won the IIT Indoor 29 Card Championship and IIT Futsal Championship in 2025.
- Represented the school football team in the Inter-Cantonment Football Championship in 2019.
- Represented Fazlul Huq House Football Team in the school House Cup Football Tournament and achieved runner-up position as a team in 2019.
- Achieved 3rd place in a school essay competition in 2017.

Languages

Bangla
English

References

Dr. Ahmedul Kabir
Associate Professor
Institute of Information Technology
University of Dhaka
kabir@iit.du.ac.bd

Dr. Rezvi Shahariar
Associate Professor
Institute of Information Technology
University of Dhaka
rezvi@iit.du.ac.bd